Project Strategic Insights & Recommendations

Risk Analytics and Claims Intelligence at ***AegisLife Insurance Pvt. Ltd.***

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Executive Report: Insurance Claims & Policy Performance Analysis

Project Overview

This project analyzed insurance policy, customer, and claims data using SQL, Python, and Power BI to identify operational inefficiencies, claim trends, and business opportunities. The analysis focused on claim outcomes, premium collection, customer segments, regional performance, and policy profitability.

A total of 1,407 claims and 1,648 customers were analyzed across multiple insurance products and geographic regions.

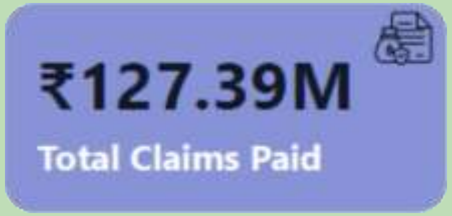
Top 5 Data-Driven Insights

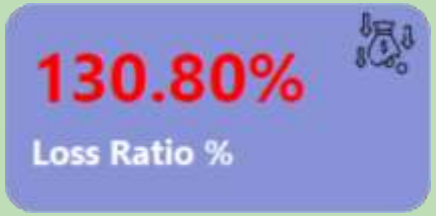
Insight 1 : Claims Outpaced Premium Revenue

Total claims paid reached **₹127.39M**, exceeding premium collected of **₹97.39M**, resulting in a **loss ratio of 130.8%**.

Business Impact : A loss ratio above 100% indicates that claim payouts exceed premium income, raising concerns about portfolio profitability and long-term sustainability.

Visualization :





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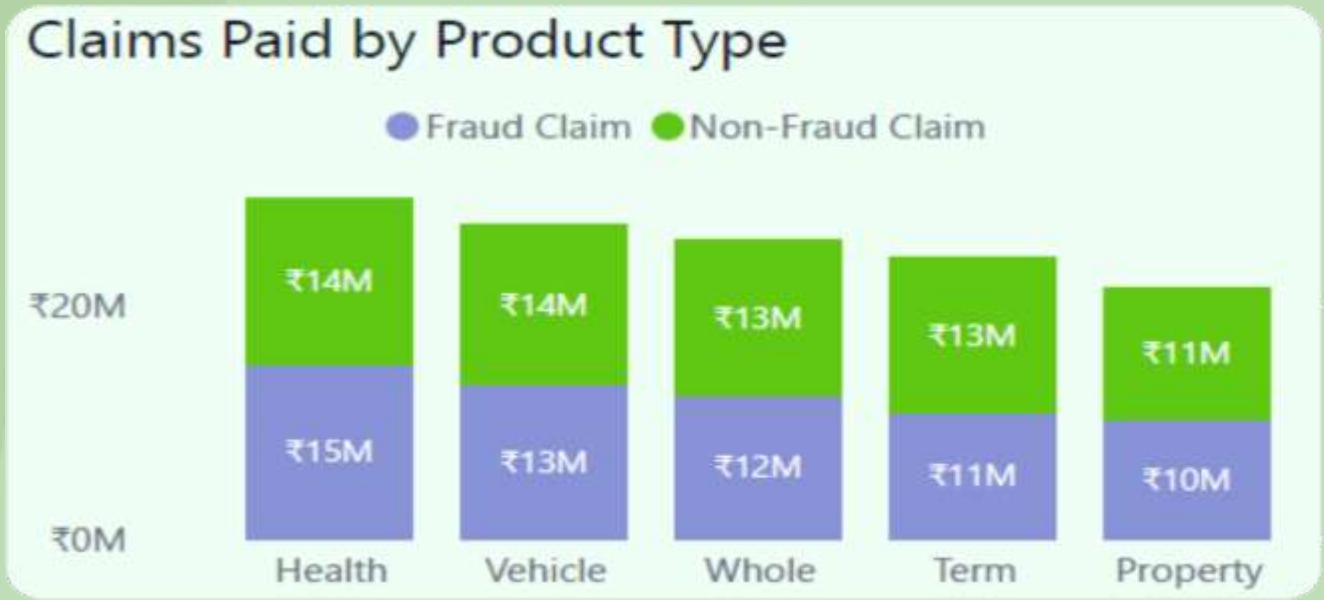
Insight 2 : Health Insurance Drive Claim Costs

Health insurance product recorded the highest claim expenses across the portfolio.

Business Impact: These product line may be underpriced or exposed to higher risk, requiring a review of underwriting guidelines and pricing strategies.

Visualization :



Insight 3 : South Region Shows Highest Financial Risk

The **South region** reported the highest loss ratio at **159.29%**, significantly above the company average.

Business Impact: Elevated claim costs in this region are reducing profitability and may require stricter risk assessment or premium adjustments.

Visualization :

Insight 4 : North Region Exhibits Elevated Fraud Risk

The **North region** recorded the highest fraud ratio at **53.25%**, indicating greater exposure to suspicious claims.

Business Impact: Concentrated fraud activity can increase claim leakage and operational costs if not proactively monitored.

Visualization :

Insight 5 : Nearly Half of All Claims Were Flagged as Fraud

The portfolio recorded a **fraud ratio of 48.97%**, indicating that nearly one out of every two claims was identified as potentially fraudulent.

Business Impact: High fraud exposure increases claim leakage, operational costs, and profitability risk. Strengthening fraud detection mechanisms can significantly improve financial performance.

Visualization :

Strategic Recommendations

Recommendation 1 : Strengthen Underwriting for Health Product

Review policy pricing, eligibility criteria, and coverage limits for high-claim products.

Expected Impact:

- Reduce claim payouts by **5–10%**
- Improve product profitability
- Lower portfolio loss ratio

Recommendation 2 : Enhance Fraud Monitoring in High-Risk Regions

Deploy advanced fraud detection models and targeted investigations in the North region.

Expected Impact:

- Reduce fraudulent claim exposure
- Improve operational efficiency
- Increase claim processing accuracy

Recommendation 3 : Implement Risk-Based Pricing

Introduce differentiated premium pricing for customers based on risk scores.

Expected Impact:

- Improve underwriting accuracy
- Increase premium revenue
- Enhance overall profitability

Recommendation 4 : Improve Agent Performance Through Training

Provide coaching and performance monitoring for agents with low productivity or high claim rejection rates.

Expected Impact:

- Increase policy sales
- Improve customer experience
- Strengthen premium generation

Recommendation 5 : Leverage Forecasting for Reserve Planning

Use predictive analytics to estimate future claim volumes and reserve requirements.

Expected Impact:

- Improve cash-flow forecasting
- Optimize reserve allocation
- Support better financial planning

Recommendation 6 : Promote Low-Risk Customer Acquisition

Prioritize marketing efforts toward low-risk customer segments while maintaining balanced portfolio diversification.

Expected Impact:

- Lower overall claim frequency
- Improve portfolio profitability
- Reduce future loss ratios
- Enhance long-term business sustainability

THANK YOU